

OpenFOAM

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Introduction to OpenFOAM

● Summary:

- Abbreviation of **Open** Field **O**peration **A**nd **M**anipulation.
- Open-source package (C++) released under the GNU General Public License (GPL).
- Based on the **cell centered Finite Volume Method**.
- Extensive **CFD** and **Multi-physics** capabilities.
- **Free** to download, install, use, modify, and distribute from the GNU GPL v3.

● OpenFOAM forks:



(official standard release)

Released Version*: 13 | **12** | 11 | 10 | 9 | 8 |
7 | 6 | 5.0 | 4.1 | 4.0 | 3.0.1 | 3.0.0 | 2.4.0 |
2.3.1 | 2.3.0 | 2.2.2 | 2.2.1 | 2.2.0 | 2.1.1 |
2.1.0 | 2.0.1 | 2.0.0 | 1.7.1 | 1.7.0 | 1.6 |
1.5 | 1.4.1 | 1.3 | 1.2 | 1.1 | 1.0

<https://openfoam.org>



(official development release)

Released Version: v2306 | v2212 | v2206 |
v2112 | v2106 | v2012 | v2006 | v1912 |
v1906 | v1812 | v1806 | v1712 | v1706 |
v1612+ | v1606+ | v3.0+

<https://www.openfoam.com>

foam-extend

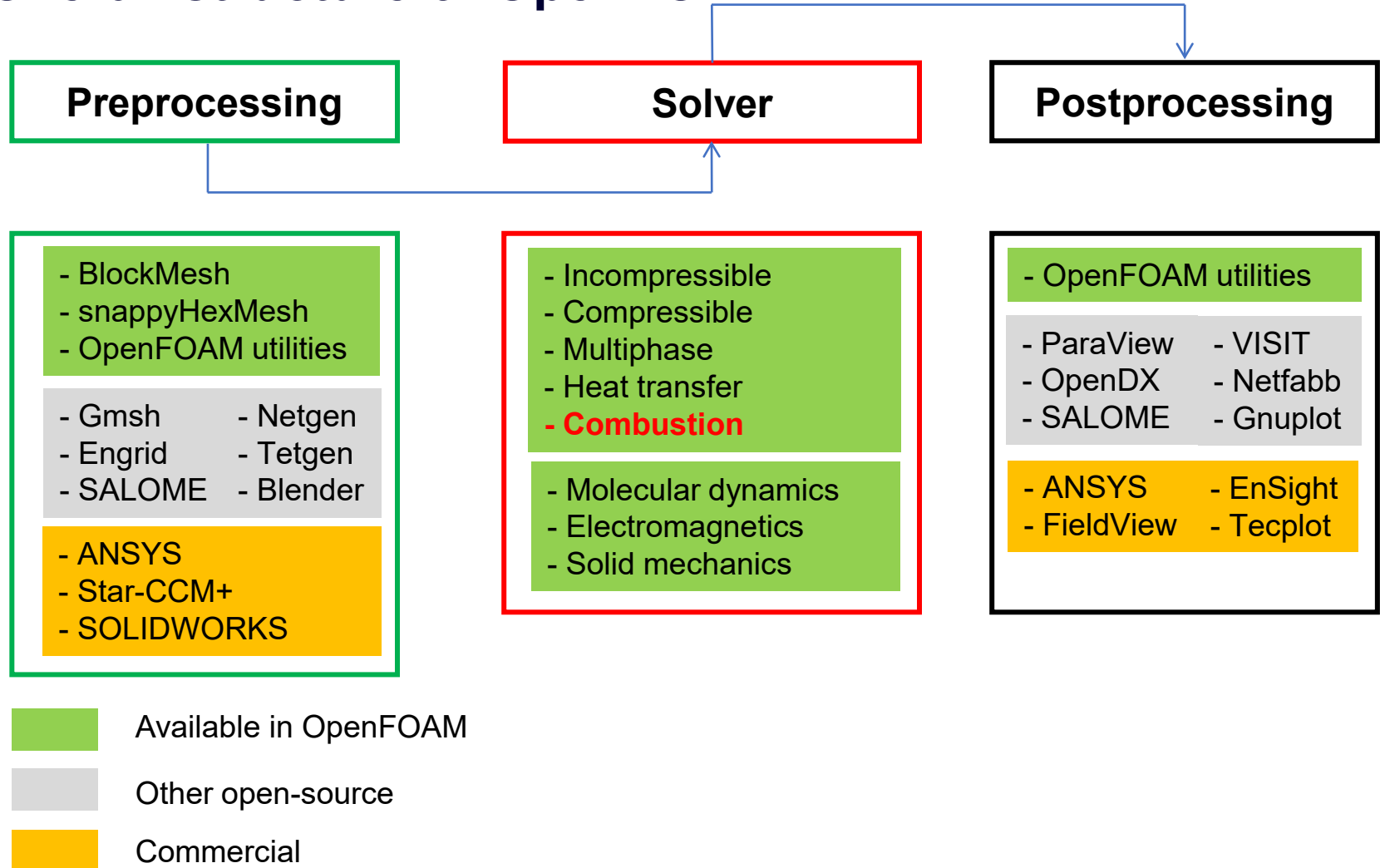
(community driven
development branch)

Released Version:
4.1 | 4.0 | 3.2 | 3.1 | 3.0

<https://sourceforge.net/projects/foam-extend>

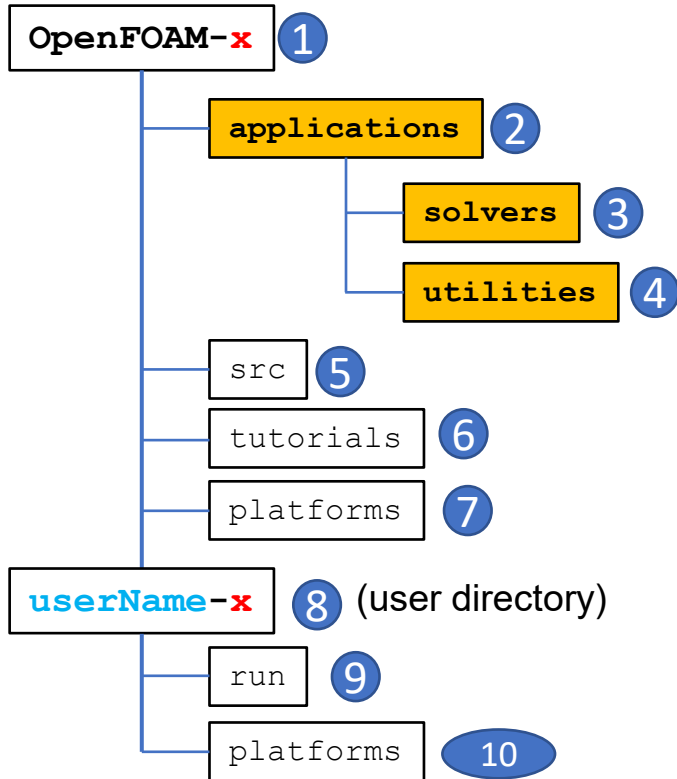
Introduction to OpenFOAM

● Overall structure of OpenFOAM



Introduction to OpenFOAM

- Code structure for Old OF version (~OF 10)



applications:

This directory contains the source files of all the executables created using C++ libraries. It contains the following directories:

- solvers:

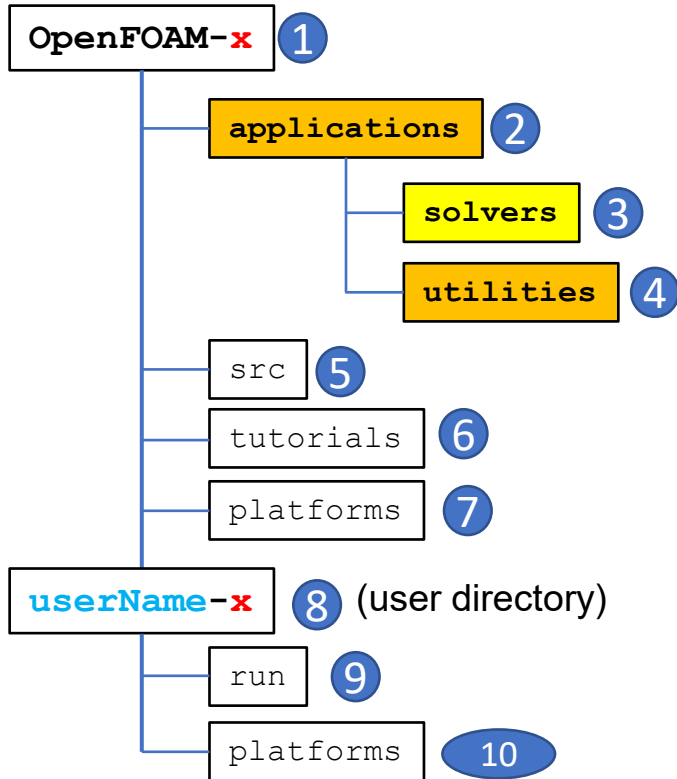
Source code to solve a particular continuum mechanics problem.

- utilities:

Source code to perform pre- and post-processing tasks involving data and algebraic manipulations.

Introduction to OpenFOAM

- Code structure for Old OF version (~OF 10)



solvers:

- compressible
- incompressible
- heatTransfer
- combustion
 - chemFoam
 - fireFoam
 - **reactingFoam**
 - **rhoReactingFoam**
 - **rhoReactingBuoyantFoam**
 - XiFoam
 - PDFFoam

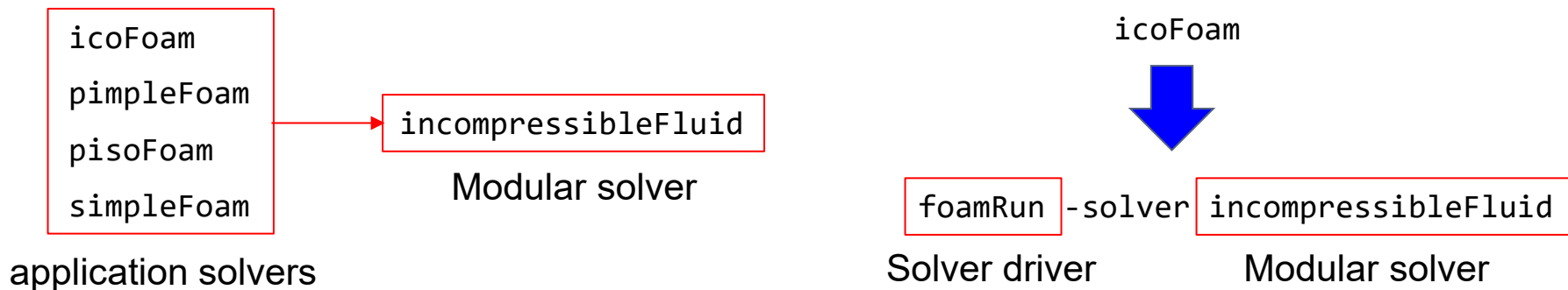
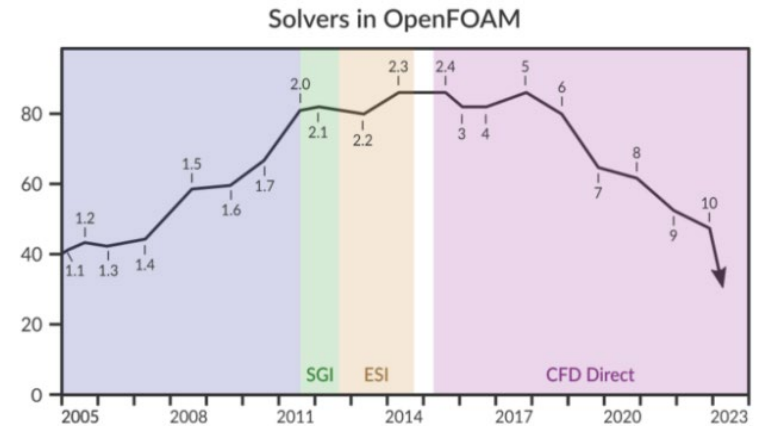
...

...

Introduction to OpenFOAM

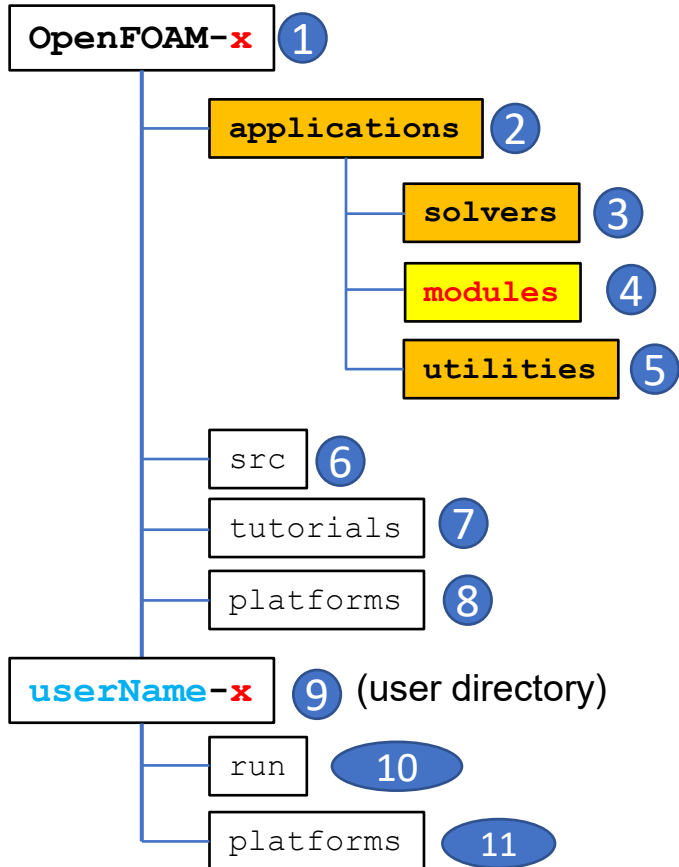
- Limitation of Old OF version (~ OF 10)

- Rapid increase in specialized application solvers
- Transition to a **modular solver** framework
- Solution strategy retained in the solver driver (solvers)
- Governing equations and physics encapsulated in modular solver classes (modules)



Introduction to OpenFOAM

- Code structure for new OF version (OF 11 – 14)



solvers:

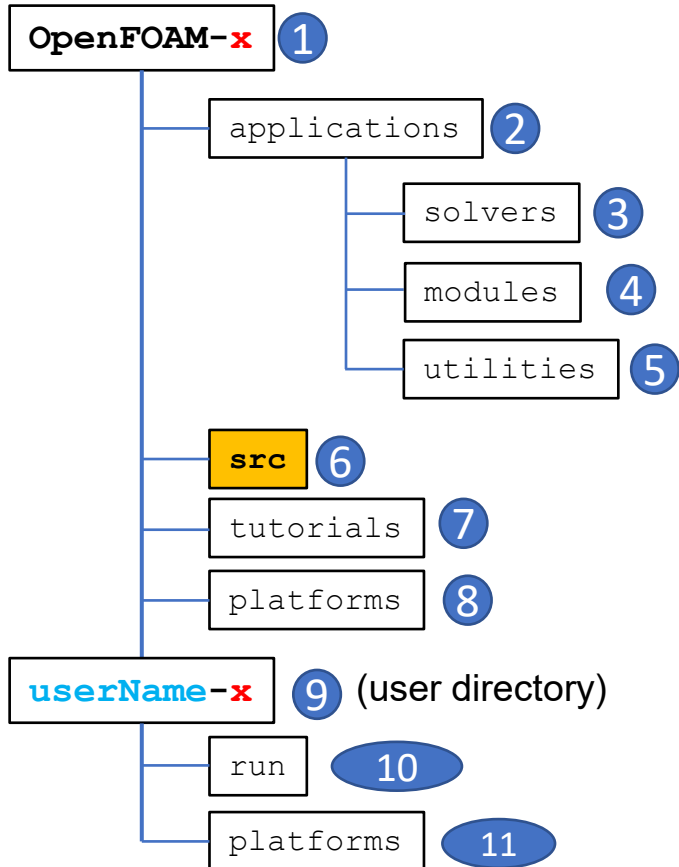
- chemFoam
- foamRun
- foamMultiRun
- ...

modules:

- incompressibleFluid
- multicomponentFluid
- ...

Introduction to OpenFOAM

- Code structure for new OF version (OF 11 – 14)



src:

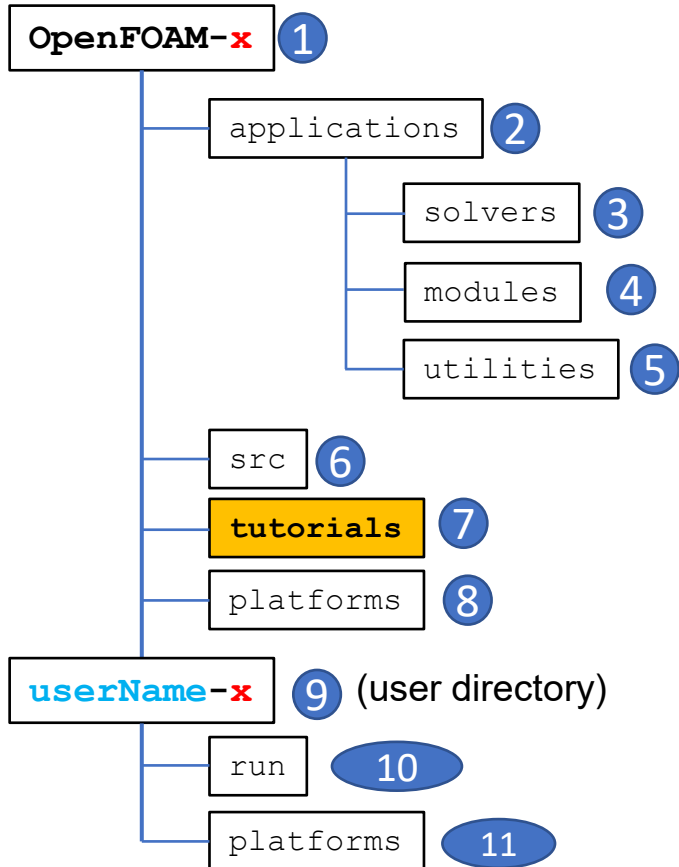
This directory contains the **source code** of all the object shared libraries. It contains the following directories (differ from ver. by ver.):

- mesh
- finiteVolume
- ODE
- combustionModels
- thermophysicalModels
- MomentumTransportModels
- ThermophysicalTransportModels

...

Introduction to OpenFOAM

- Code structure for new OF version (OF 11 – 14)

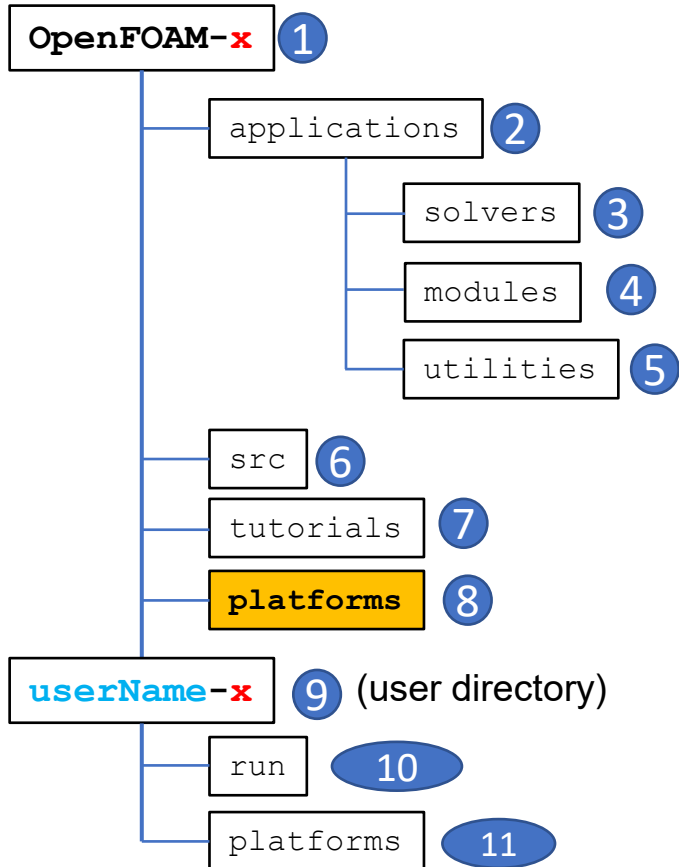


tutorials:

This directory contains the running case files as examples w.r.t **modules** in the applications directory for continuum mechanics problems:

Introduction to OpenFOAM

- Code structure for new OF version (OF 11 – 14)

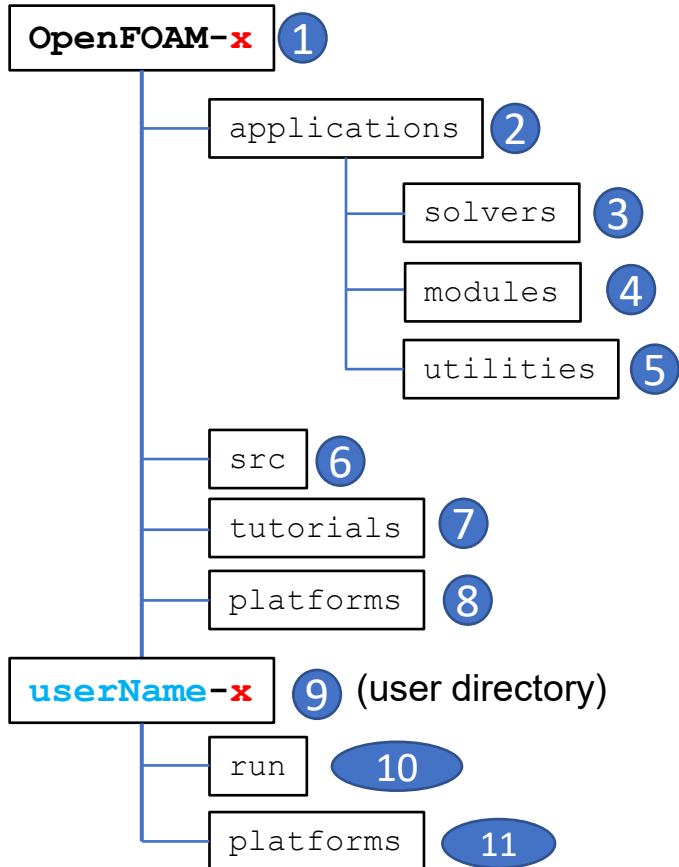


platforms:

This directory contains the binary files such as object and executable files:

Introduction to OpenFOAM

- System variables and aliases:



(main) Variable	Alias	Go to directory
\$WM_PROJECT_DIR	foam	1
\$FOAM_APP	app	2
\$FOAM_SOLVERS	sol	3
\$FOAM_MODULES	mod	4
\$FOAM_UTILITIES	util	5
\$FOAM_SRC	src	6
\$FOAM_TUTORIALS	tut	7
\$FOAM_LIBBIN	lib	8
\$FOAM_APPBIN		8
\$WM_PROJECT_USER_DIR		9
\$FOAM_RUN	run	10
\$FOAM_USER_LIBBIN		11
\$FOAM_USER_APPBIN		11